

Networking > Networking Software > ... >

mlx_fpga – Burning and Debugging Tool for NVIDIA Devices with FPGA

mlx_fpga – Burning and Debugging Tool for NVIDIA Devices with FPGA

The mlx_fpga utility will be deprecated as of MFT v4.18.0.

mlx_fpga tool allows the user to burn and update a new FPGA image on NVIDIA Innova™ adapter cards. For instructions on how to burn, please refer to [Burning the FPGA's Flash Device using the mlx_fpga Burning Tool](#).

The tool also enables the user to read/write individual registers in the FPGA configuration space.

Tool Requirements

- An Innova IPsec 4 Lx EN / Innova Flex 4 Lx EN adapter card with an FPGA device
- For Innova IPsec 4 Lx EN: Load the mlx5_fpga_tools module
- For Innova Flex 4 Lx EN: Load the mlx_accel_tools module

Note: The module is included in the Innova driver which is supplied for this product line only, and available through the [Support](#).

- Start mst service with the fpga lookup flag (mst start --with_fpga)

mlx_fpga Synopsis

where:



-d --device <device>	FPGA mst device interface
-v --version	Display version info
-h --help	Display help message
-f --force	Non-interactive mode, answer yes to all questions
r read <addr>	Read debug register in address

w write <addr> <data>	Write data to debug register in address
b burn <image file/s>	Burn the image on the flash
l load	Load image from flash (--factory - load image from factory flash)
clear_semaphore	Unlock flash controller semaphore
reset	Reset flash controller (--gw) or FPGA (--fpga)
q query	Query general FPGA information
set_fw_mgmt <disable enable>	Disable/Enable FPGA management by the Firmware

Examples of mlx_fpga Usage

Adding FPGA mst Device Interface

For Innova IPsec 4 Lx EN: Load the mlx5_fpga_tools module.

```
1  # modprobe mlx5_fpga_tools
2  # mst start --with_fpga
3  # mst status
4  MST modules:
5  -----
6  MST PCI module is not loaded
7  MST PCI configuration module is not loaded
8  MST devices:
9  -----
10 No MST devices were found nor MST modules were loaded.
11 You may need to run 'mst start' to load MST modules.
12 FPGA devices:
13 -----
14 /dev/mst/mt4117_pciconf0_fpga_i2c
15 /dev/mst/mt4117_pciconf1_fpga_rdma1
```

Note: In the last line, it is recommended to use the RDMA device as it is faster. I2C is used for recovery purposes when RDMA is not functional.

For Innova Flex 4 Lx EN: Load the mlx_accel_tools module.

```
# modprobe mlx_accel_tools
# mst start --with_fpga
# mst status
MST modules:
-----
MST PCI module is not loaded
```

```

8   MST PCI configuration module is not loaded
9   MST devices:
10  -----
11  No MST devices were found nor MST modules were loaded.
12  You may need to run 'mst start' to load MST modules.
13  FPGA devices:
14  -----
15  /dev/mst/mt4117_pciconf0_fpga_i2c
    /dev/mst/mt4117_pciconf1_fpga_rdma

```

Note: In the last line, it is recommended to use the RDMA device as it is faster. I2C is used for recovery purposes when RDMA is not functional.

For recovery only, where modules are broken or missing.

```

1   # mst start --with_fpga_fw_access
2   Starting MST (Mellanox Software Tools) driver set
3   Loading MST PCI module - Success
4   Loading MST PCI configuration module - Success
5   Create devices
6   Unloading MST PCI module (unused) - Success
7   # mst status
8   MST modules:
9   -----
10  MST PCI module is not loaded
11  MST PCI configuration module loaded
12  MST devices:
13  -----
14  /dev/mst/mt4117_pciconf0          - PCI configuration cycles access.
15                                     domain:bus:dev.fn=0000:01:00.0 addr.reg=8
16                                     Chip revision is: 00
17  FPGA devices:
18  -----
19  /dev/mst/mt4117_pciconf0_fpga

```

The `mlx5_fpga_tools` and the `mlx_accel_tools` modules must be down before trying to recover the FPGA mst Device Interface.

Burning the FPGA's Flash Device using the `mlx_fpga` Burning Tool

`mlx_fpga` tool burns a .bin file onto the FPGA flash device.

The Innova Flex .bin file must be generated according to the instructions listed in the *Innova Flex Adapter Card User Manual*.

It is recommended to burn the FPGA device using an RDMA device as it faster and it shortens the burning time.

1. Burn the image.

```
1 # mlx_fpga -d <device> burn image.bin
```

2. Load the FPGA image from flash according to “Loading the Tool” below or power cycle the machine for change to take effect.

Loading the Tool

Load an FPGA image from user configurable flash:

```
1 # mlx_fpga -d <device> l/load <optional: load options>
```

where <optional: load options> is:



--factory	Load FPGA image from factory flash
--user	Load FPGA image from user flash [Default option]

Debugging the Tool

Reading One Debug Register:

```
1 # mlx_fpga -d <device> read 0x0
```

Writing One Debug Register:

```
1 # mlx_fpga -d <device> write 0x0 0x0
```

Updating the FPGA Image

In order to verify the new image burned to the FPGA, the user can use mlx_fpga tool and read the following registers:



Name	Address	Range	Default
image_version	0x900000	31:00:00	0x0

image_date	0x900004	31:00:00	0x0
image_time	0x900008	31:00:00	0x0

Innova Flex users should refer to the Innova User Manual for more information.



Previous page

[mlxphyburn – Burning Tool for
Externally Managed PHY](#)

Next page

[cpldupdate – Tool for
Programming On-Board CPLDs
on NVIDIA Devices](#)



 Use with AI 



[Privacy Policy](#)

[Your Privacy Choices](#)

[Terms of Service](#)

[Accessibility](#)

[Corporate Policies](#)

[Product Security](#)

[Contact](#)

Copyright © 2026 • Powered by Scroll Sites & Atlassian Confluence